

Gene Pulser® Electroporation

Cell Type Bacterial, gram negative

Molecules Electroporated DNA: plasmid from patient isolate

Species Used *E. coli*, K12 (derivatives)

Before the Pulse

Cell Growth Medium L-broth

Growth Phase at Harvest O.D. (600) =0.35

Pre-pulse Incubation 0.5 to 1 min.

Wash Solution water

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 0 °C

Electroporation Medium 10% glycerol

Cuvette Gap 0.2 cm

Cell Density 2.7 x 10 (9) cells / µl

Voltage 2.5 kV

Volume of Cells 40 µl

Field Strength 12.5 kV/cm

DNA Concentration about 30 ng / µl

DNA Resuspension Buffer TE buffer (10 mM Tris, 1 mM EDTA, pH 8.0)

Capacitor 25 µF

Volume of DNA 2 µl

Resistor 200 Ω (Pulse Controller)

After the Pulse

Time Constant 4.6 msec

Outgrowth Medium SOC - medium

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
SOC: 2% Bacto tryptone, 0.5% Bacto yeast extract, 10mM NaCl, 2.5mM KCl, 10 mM MgCl₂, 10 mM MgSO₄, 20 mM glucose.
LB: 1% Bacto tryptone, 0.5% Bacto yeast extract, 0.5% NaCl.

Outgrowth Temperature 37 °C

Length of Incubation 1 hr.

Selection Method or Assay Used Nalidixin, Rifampicin, Cefotaxime

Electroporation Efficiency 6 x 10 (5) transformants/ µg DNA

Per Cent Survival Not given

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Survey Number
 033